MIL-W-80164A
26 July 1983
SUPERSEDING
MIL-W-80164
12 September 1968

MILITARY SPECIFICATION

WELDING HEADS, ARC, AUTOMATIC, CONSUMABLE WIRE ELECTRODE

"This specification is approved for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 Scope. This specification covers automatic welding heads capable of performing electric arc welding operations with or without shielding gas, utilizing consumable wire electrodes. The welding power supply is not included in this specification.

1.2 Classification. The welding heads covered by this specification shall be of the sizes listed in table I. The size required shall be as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

*2.1.1 Specification and standard. Unless otherwise specified the following specification and standard of the issue in effect on the date of the invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, TN 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following other government documents form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IEEE 268 - Metric Practice

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of conflict between the test of this specification and the references cited herein, the test of the specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1 and 6.3), the supplier shall furnish one complete welding head of each size specified in the contract or order for first article inspection in accordance with section 4 herein.

* 3.2 Design. The welding heads shall be of the manufacturer's standard commercial product, shall be new and include any additional features necessary to comply with the requirements specified herein. Additional features which are not specified herein that are a part of the manufacturer's standard product shall be included in the equipment furnished. Design of the welding heads shall provide for controlling the entire welding operation including

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electrode feeds, welding speeds and interruptions to provide welding processes specified herein. The heads shall be as compact and lightweight as possible, consistent with good commercial design practice. The welding head design and electrical system shall be compatible to the manufacturer's power supply.

3.2.1 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding heads that present safety hazards. The safety devices shall not interfere with operation of the welding head. The safety devices shall prevent unintentional contact with the guarded part and shall be removable only with tools to facilitate inspection, maintenance, and repair for the parts. All parts, components, mechanisms and assemblies furnished on the welding head, whether or not specifically required herein, shall comply with all applicable requirements of OSHA 2206. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.2 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of parts or welding head.

3.2.3 Materials. Materials not specified herein shall be selected by the manufacturer to meet the requirement of this specification. Contact between dissimilar metals, especially brass, copper or steel in contact with aluminum or magnesium shall be avoided whenever possible. Materials shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

3.2.4 Reclaimed materials. The welding heads may contain reclaimed materials provided such materials will not jeopardize the intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding head.

3.2.5 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the welding head. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.6 Lubrication. All bearings (except the sealed-for-life type) and moving parts shall be provided with means for lubrication. All lubrication systems shall have means for adding or changing lubricant and shall have means for sealing all openings to prevent dirt and foreign matter from entering. Grease and oil fittings shall be easily accessible. The supplier shall furnish all lubricants required for the initial operation of the equipment.

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*3.3 Construction. The welding heads shall be constructed so that when installed and connected to the power sources, they can be used for any purpose specified herein. The welding heads shall be constructed of parts which are new, without defects and free of repair.

3.3.1 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.2 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Painting. The welding heads shall be painted in accordance with the manufacturer's commercial practice.

3.3.5 Threads. All threaded parts shall be in accordance with FED-STD-H-28.

3.3.6 Gears. All gears shall be of a hardness and toughness that will enable the gear system to transmit full rated torque of the drive motor without gear damage, or premature wear.

3.4 Electrical.

*3.4.1 Control input power. Unless otherwise specified (see 6.2.1), the welding heads shall operate on an independent nominal power input of 120 volts, 60 Hz power supply.

3.4.2 Circuitry. The type and size of electrical wiring used in all circuits shall have sufficient capacity, temperature, voltage, and insulation rating for the intended use. Wiring subject to chafing, vibration, and tension shall be so installed to prevent damage. The wire and terminals shall be color or number coded.

3.5 Performance. When connected to the proper power sources and operated in accordance with the manufacturers printed instructions for automatic welding applications, the welding heads shall automatically feed consumable wire electrodes and produce satisfactory welds on ferrous and non-ferrous metals and alloys of metals within the capacities and requirements speci-

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fied herein. The welding heads shall be capable of operating continuously at rated output without exceeding the designed operated conditions of any of the components. The welding heads shall be capable of utilizing and controlling the range of input currents of the types as shown to effectively perform arc welding with the sizes of wires at the range of wire speeds as shown in table I. The welding heads shall be capable of welding under any of the following processes:

a. Metal insert gas shielded welding using argon or helium.

b. Metal gas shielded welding using CO_2 or CO_2 and argon.

c. Tubular flux cored wire welding, with or without CO_2.

TABLE I

Size and characteristics

Size	Ampere Rating	Electrode Holder			*Welding Head, Wire Feed Rate, Inches Per Minute, ± 20%
		Wire Size, Diameter, Inches			
		Hard	Soft	Cored	
1	350	.030-1/16	.030-5/64		50-600
2	500	.030-1/16	.030-1/8	1/16-1/8	50-600
3	600	.030-1/8	.030-1/8	1/16-1/8	30-300
4	700	.030-1/8	.030-1/8	1/16-1/8	90-900
5	750	.030-1/8	.030-1/8	1/16-1/8	25-400
6	800	.030-1/8	.030-1/8	5/64-1/8	50-155

3.6 Components.

3.6.1 Control unit. The control unit for the welding head shall provide the necessary degree of control to assure that all components of the welding head operate properly and in the desired sequence. Wire connections shall be of the screw, snap on, or spring clip type.

3.6.1.1 Solenoid gas valve. The electric control unit shall have a solenoid valve to start and stop the flow of the shielding gas. The gas shall start and stop simultaneously with the welding operation. When specified (see 6.2.1), suitable timers shall be provided to permit preflow and post-flow of the shielding gas.

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* 3.6.1.2 Liquid coolant valve. When water cooled electrode holders are to be used in conjunction with a standard water tap supply, a solenoid valve shall be furnished to control the flow of coolant through the holder and the valve shall be designed to operate at pressures delivered at a standard water tap supply. The valve may operate in conjunction with the gas valve specified in 3.6.1.1. Means shall be offered as an option for automatically shutting down the welding operation in event of failure in the flow of liquid coolant.

3.6.2 Wire feeder unit. The wire feeder unit, when used in combination with various accessory equipment, shall automatically feed the welding wire to the weld zone at controlled rates. The feeder shall consist basically of a wire feed motor, gear reduction unit, wire drive, contact devices, accessory support and mounting clamp, and accessories. The wire feeder unit shall be compatible with the control unit, and with the welding power as supplied by the proper welding power supply. Wire feed rate accuracy shall be within $\pm$5% of preset rate with line voltage fluctuation of up to $\pm$10%. With the line voltage constant, the feed rate shall be within $\pm$2% of preset wire feed rate. The unit shall have some degree of flexibility allowing the unit or its necessary components to be raised or lowered, moved laterally, or swiveled to meet various welding applications. The wire feeder unit shall include the necessary devices to assure that the feed of the wire will stop when the wire feed switch is put in the "stop" position.

3.6.2.1 Wire feed motor. The motor shall be of adequate size and design to drive the wire feed at all speeds and conditions specified herein. The motor shall have sealed for life ball or roller bearings and be so designed to allow brush removal and replacement without disassembling the motor.

3.6.2.2 Gear reduction unit. The gear reduction unit shall be driven by the wire feed motor and shall reduce the motor output shaft speed to the desired speed.

*3.6.2.3 Wire drive. The wire drive unit shall be driven by the gear reduction unit and shall consist of all necessary devices to properly move the wire. The drive shall consist of one or more feed rolls and pressure rolls. The feed rolls shall be of the friction type and grooved as required to grip and drive the wire, and the groove shall be of such size to fit the wire in use and not to deform it to such a degree to affect the welding arc or to cause excess friction or wear. The wire feed unit shall be capable of feeding all sizes of wire specified in table I by manual shifting of gears or by changing sizes of gears or rolls. The pressure rolls shall provide the necessary positive feeding force to effectively feed the wires listed in table I.

3.6.2.4 Wire reel mount. The welding system shall have provisions for mounting commercially available wire reels.

*3.6.2.5 Wire straightener. When required, welding heads shall be furnished with a wire straightening device. The straightener shall consist of rolls or similar devices to remove kinks and bends from the welding wire before the wire enters the wire feeder. The rolls, or similar devices, shall be hardened to resist wear, shall be flexible to allow for different positioning of the straightener and shall handle the applicable sizes and types of wire specified (see 6.2.1).

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3.6.3 Enclosure. All components of the welding head shall be suitably enclosed or protected. The enclosures as required shall be of such construction to support and protect the contents. Adequate means such as doors or access panels shall be provided to assure easy access for servicing.

3.6.4 Interconnecting cables and hoses. The welding heads shall have all necessary electric codes and cables, and hoses for the proper interconnection of all components. All cords and cables shall be adequately fastened to withstand normal strains and flexibly mounted to accommodate all movements of the various components. Hoses shall be of the abrasion resistant type, properly fastened to withstand normal strains, and flexibly mounted to accommodate the normal changing positions utilized. The sizes and lengths of all cables and holes shall meet all requirements of this specification. When required to meet special conditions, the lengths of all cables and hoses shall be as specified (see 6.2.1).

3.6.5 Gas flowmeter and regulator. Each welding head shall have a flowmeter and regulator to control the flow of shielding gas to the electrode holder and to measure the cylinder pressure. Unless otherwise specified (see 6.2.1), the flowmeter shall have a scale calibrated for use with argon gas and shall have a flow rate of 0-60 CFH. The flowmeter scale shall be protected by a clear high impact resistant material, and shall be easily removed for cleaning or changing scales. The regulator shall have a gage to register the pressure in the gas cylinder.

3.6.6 Electrode holder. Contact devices shall conduct the welding power into the electrode holder. The electrode holder shall be adaptable to the welding head, and the holder shall be rated for continuous duty. The contact nozzle assembly shall transmit the welding power into the wire and shall consist of wire guides sized for the wire in use, the nozzle, and the insulating nozzle. The wire guides shall be easily replaced and a complete set of guides, as necessary to feed the sizes of wire shown in table I for the required unit specified, shall be furnished. The ampere rating and characteristics of the electrode holder shall be air or water cooled, as specified (see 6.2.1). All outer surfaces of the holder shall be electrically insulated from the welding current. The holder body shall be provided with a conduit for feeding and guiding the electrode wire. Water cooled holders shall be compatible with permanent type anti-freeze solutions.

* 3.6.6.1 Nozzle cup. The holder shall have a replaceable nozzle cup to direct the flow of shielding gas to completely shield the electrode and weld area from the surrounding atmosphere. The cup shall be designed to be replaced without the use of tools and it shall be plated or lined, or of a material to prevent excessive build-up of spatter from the arc. The quantity of nozzle cups provided with each holder shall be as specified, (see 6.2.1).

3.6.6.2 Contact tips and tubes. The holder shall have a replaceable contact tip or tube to transfer the welding current to the electrode. The tip or tube shall be easily replaced using common tools. The quantity and type of tips and tubes required shall be as specified (see 6.2.1).

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3.7 Equipment. The unit shall be furnished with all standard equipment normally furnished by the supplier including any special tools or wrenches normally furnished.

3.8 Optional equipment. When specified (see 6.2.1), optional equipment shall be furnished and installed as required. Optional equipment as required for special functions shall be completely compatible with all required components of the welding head and where possible, controlled from the control unit.

* 3.9 Nameplate. A corrosion resistant metal nameplate shall be securely attached to each head enclosure. The nameplate shall contain the information listed below. If the head is a special model, the model designation shall include the model of the basic standard unit and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code
U.S.

3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the welding equipment and accessories shall be commensurate with the requirements specified herein, and shall be of a quality prevailing among manufacturers normally producing equipment of the type specifeid herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall be applied when required in accordance with 3.1. Failure of the first article tests shall be cause for rejection. Unless otherwise specified (see 6.2.1),

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first article inspection shall consist of the examination in 4.5 and all tests in 4.6.

4.4 Quality conformance inspection. Quality conformance inspections shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5 and the inspection in 4.7. Each item shall comply with all requirements.

4.5 Examination. The welding heads shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The welding head shall be connected to the proper power sources and operated without any wire feeding through the wire feeder for a minimum of 30 minutes, during which time the controls, the wire feed motor, and associated components shall be properly adjusted, actuated and observed to determine if the welding head operates properly.

4.6.2 Performance test. Welding test shall be performed using the smallest and largest size welding wire required for the size head. The performance test shall demonstrate the capability of the head to start as required and to operate at continuous duty, utilizing the range of currents, wire speeds, the control and flow of shielding gas and liquid coolant, if applicable. The finished welds shall be inspected by the radiographic method or other non-destructive testing methods to assure that the welding heads will produce satisfactory welds.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging - packing and markings. Unless otherwise specified, packaging, packing and marking shall be in accordance with ASTM Standard D 3951. When required, level A packaging and level A or B packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. The required levels and special marking, if required shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The welding heads covered by this specification, when connected to the proper power source, and to a stationary or mechanized fixture are to be used in performing automatic arc welding of metals using the consumable wire electrode method with or without a protective atmosphere of shielding gas.

6.2 Ordering data.

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6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.

b. Size required (see 1.2).

c. First article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.1 and 6.4).

e. Measurement system, if different (see 3.2.5).

f. Power supply, if different (see 3.4.1).

g. Pre-flow and post-flow gas timer, if required (see 3.6.1.1).

h. Wire straightener, if required. (Specify size of wire) (see 3.6.2.5).

i. Cables and hose length, if different (see 3.6.4).

j. Flowmeter, if different (see 3.6.5).

k. Specify air or water cooled electrode holder, as required (see 3.6.6).

l. Specify quantity of nozzle cups required (see 3.6.6.1).

m. Specify quantity of tips and tubes required (see 3.6.6.2).

n. Optional equipment if required (see 3.8).

o. First Article inspection, if different (see 4.3).

p. Quality conformance inspection if different (see 4.4).

q. Military packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or B packing and specify special marking if required.)

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a pre-production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.1 requires compliance only with those OSHA requirements that concern "the unit in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors and air contaminants in the environment. (Does not limit the hazard level of individual units in the environment.) The procuring agency is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional unit requirements that will integrate the new welding sets into their future operating environment. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1.(d).

6.5 Revision asterisks. The margins of this specification are marked with asterisks to indicate where changes (additions, modification, corrections, and deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Navy - SH
Air Force - 99

Preparing Activity:

DLA-IP

Review Activities:

Navy - AS, OS, SH
Air Force - 84
DLA - GS

Project Number:

3431-0146

User Activities:

Navy - YD, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80164A	Military Specification (Welding Heads, Arc, Automatic, Electrod

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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